

# TEAM J8 RAINWATER HARVESTING SYSTEM

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RAINWATER HARVESTING SYSTEM**

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**OUR UNIQUE STRATEGY**

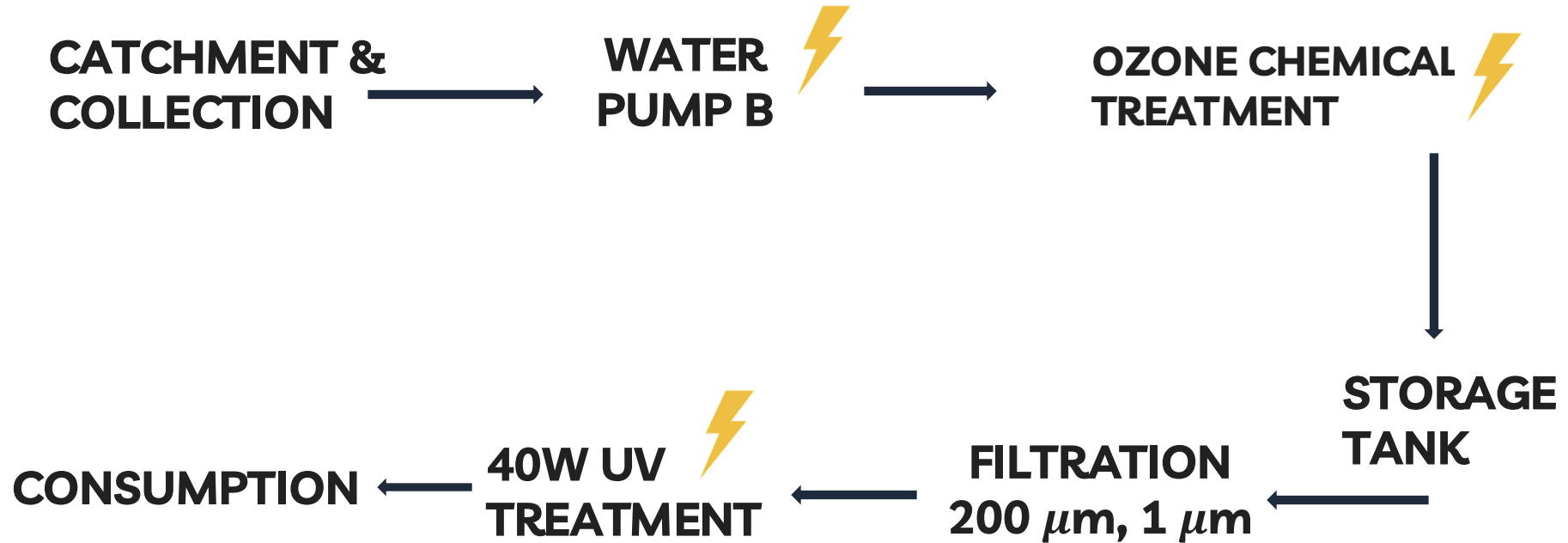
**6**

**RESULTS OF ANALYSIS**



# FINAL DESIGN ELEMENTS

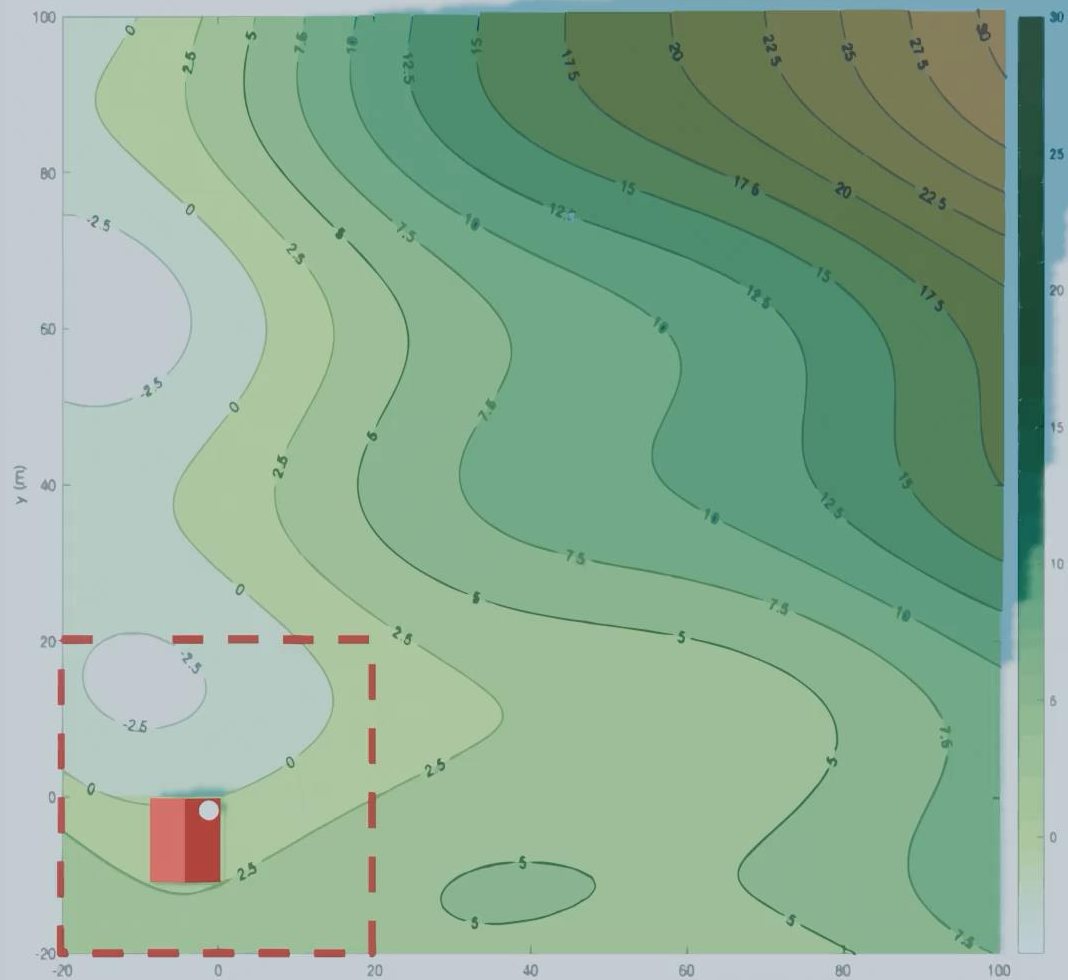
Power: Solar Panels





# Off-Grid Rainwater Harvesting System for Van Anda





# Stakeholders

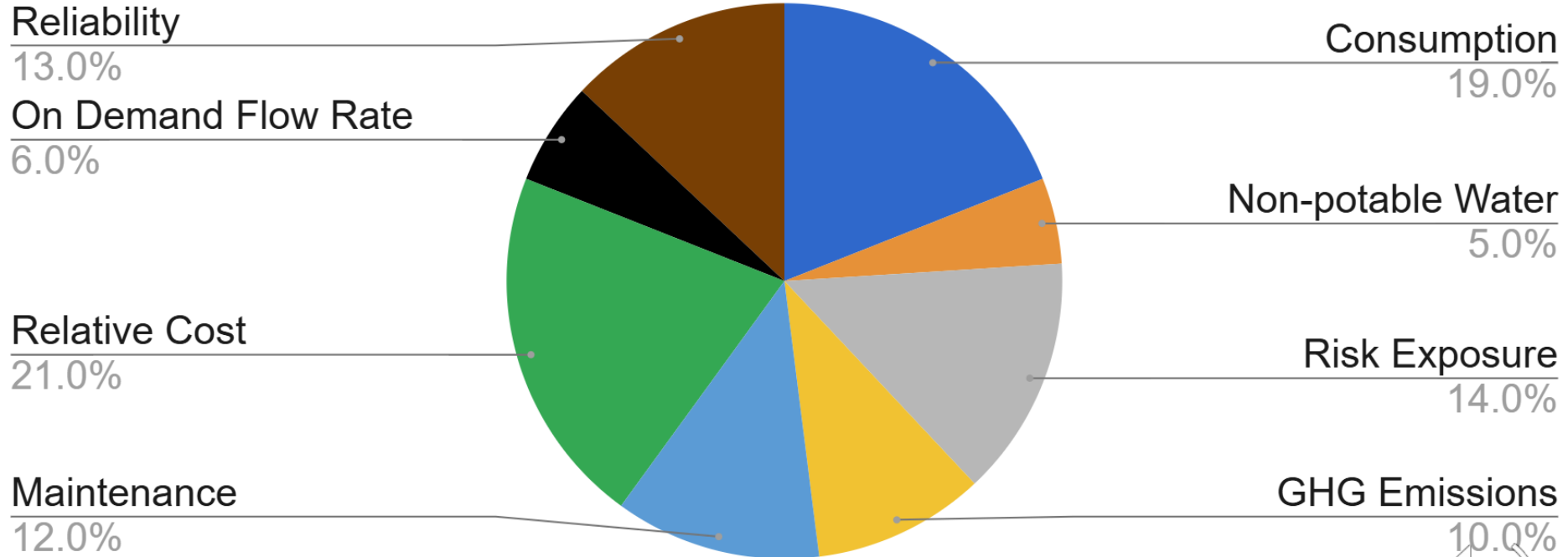
- Residents of Van Anda
- Maintenance Workers
- System Operators

## Needs:

- Low cost and environmental impact
- Meets flow rate and water supply needs
- Minimal maintenance operations
- Safety for operators



# Weighted Decision Matrix Criteria



# Pumps

Changing pump impacts satisfaction of relative cost and greenhouse gas emissions

**Pump B increases this satisfaction the most**

- 246LPM high flowrate, 1650 running hours, \$1250 initial cost



# Relative Cost

- Our system is **93.7%** less expensive than shipping in the equivalent amount of water
- Although we have more upfront cost our system provides long term savings



# Energy Source: Solar Panels

Three options: **HES-305P**, SW-80, HES-260

Final Choice: **HES-305P, 13 Panel**

**Optional balance of area and efficiency for a fair price**



# Maintenance

Our solution minimizes maintenance to 22 operations/year

- Filter Replacement
- Pump replacements
- Solar Panel Seasonal Cleaning

Our choices increases satisfaction in the Maintenance Occurrences category



# Health and Environmental Risk Exposure

## Zero risk exposure!

Solar Panels and Ozone are extremely safe for the environment and operators.

Reduced greenhouse gas emissions in compared to diesel.



# Stakeholder Satisfaction

Testing a wide variety of options for our RWH system we found the **overall satisfaction** score to be:

**8.4/10**



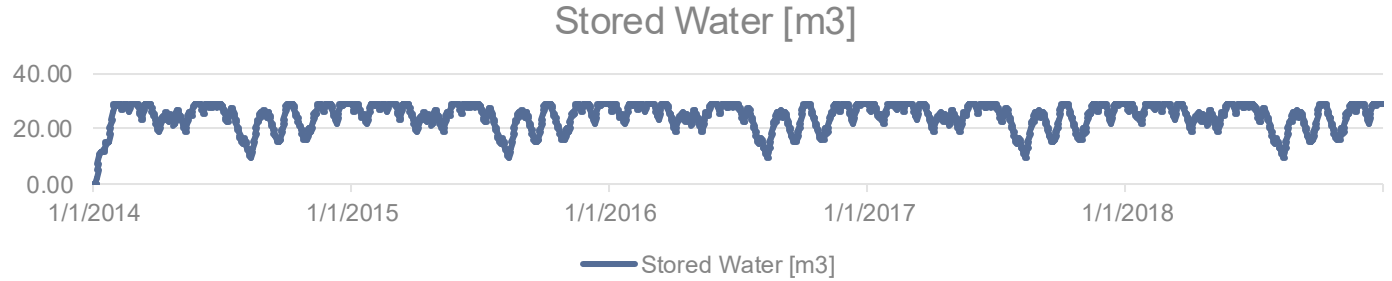
**WHAT SEPERATES  
US?**

**WE REALLY FOUND THE  
OPTIMAL SOLUTION**

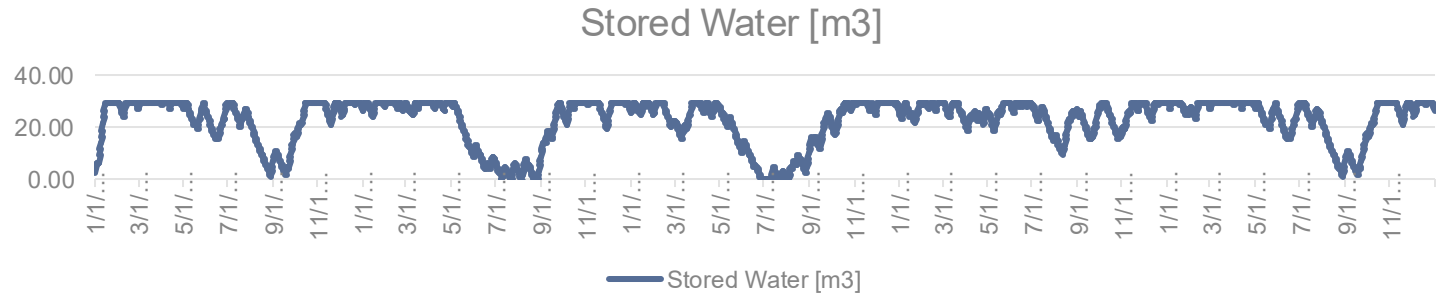


# Climate Resiliency through Rainfall Data

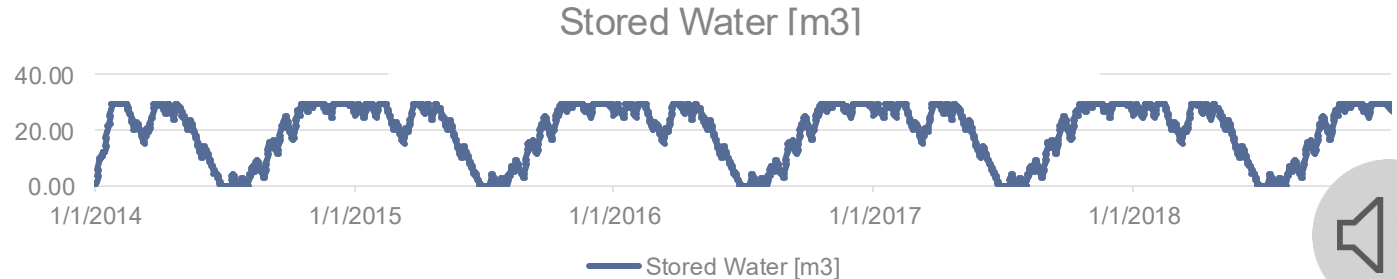
Ideal: 8.4/10



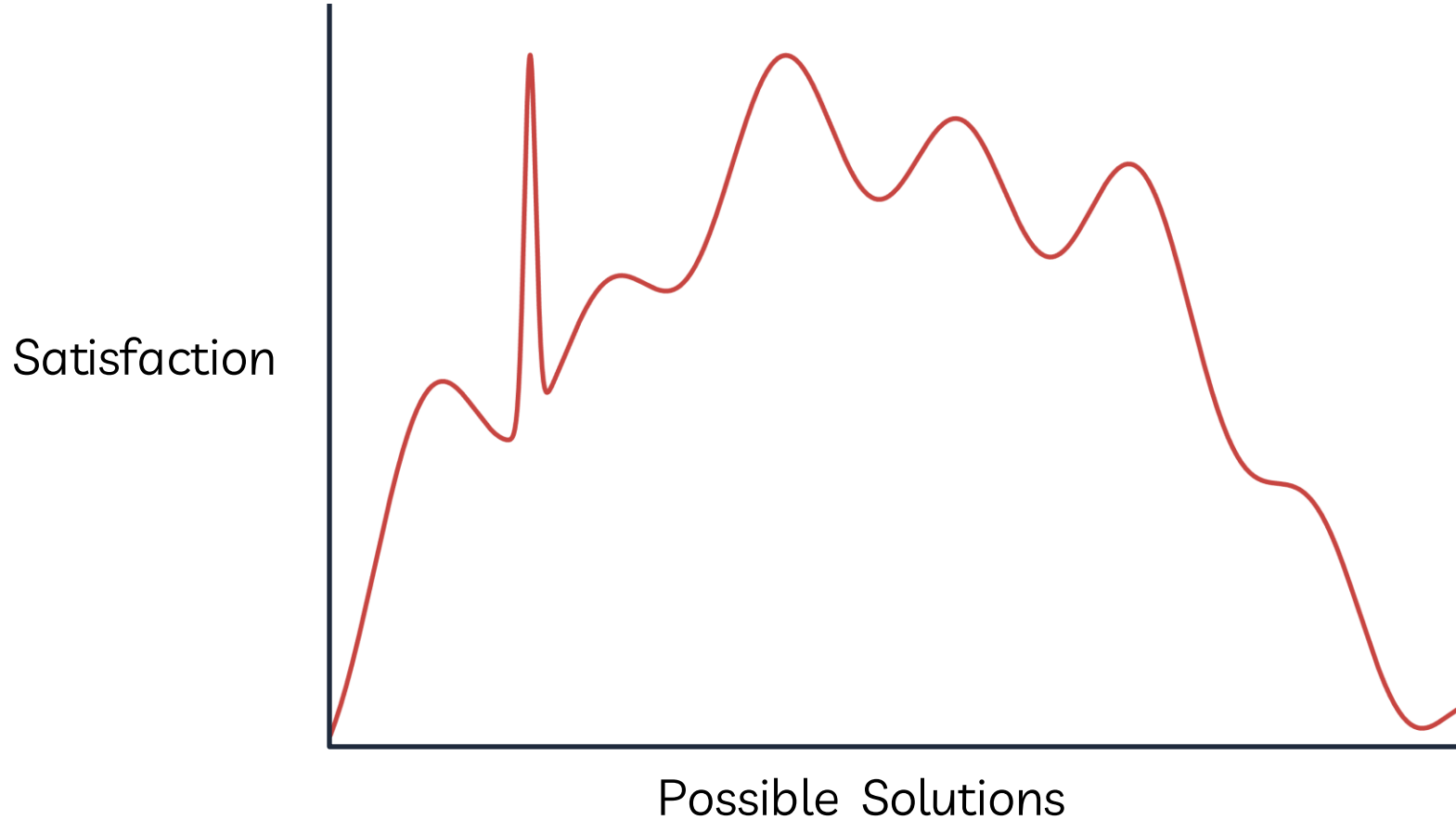
Standard: 8.2/10



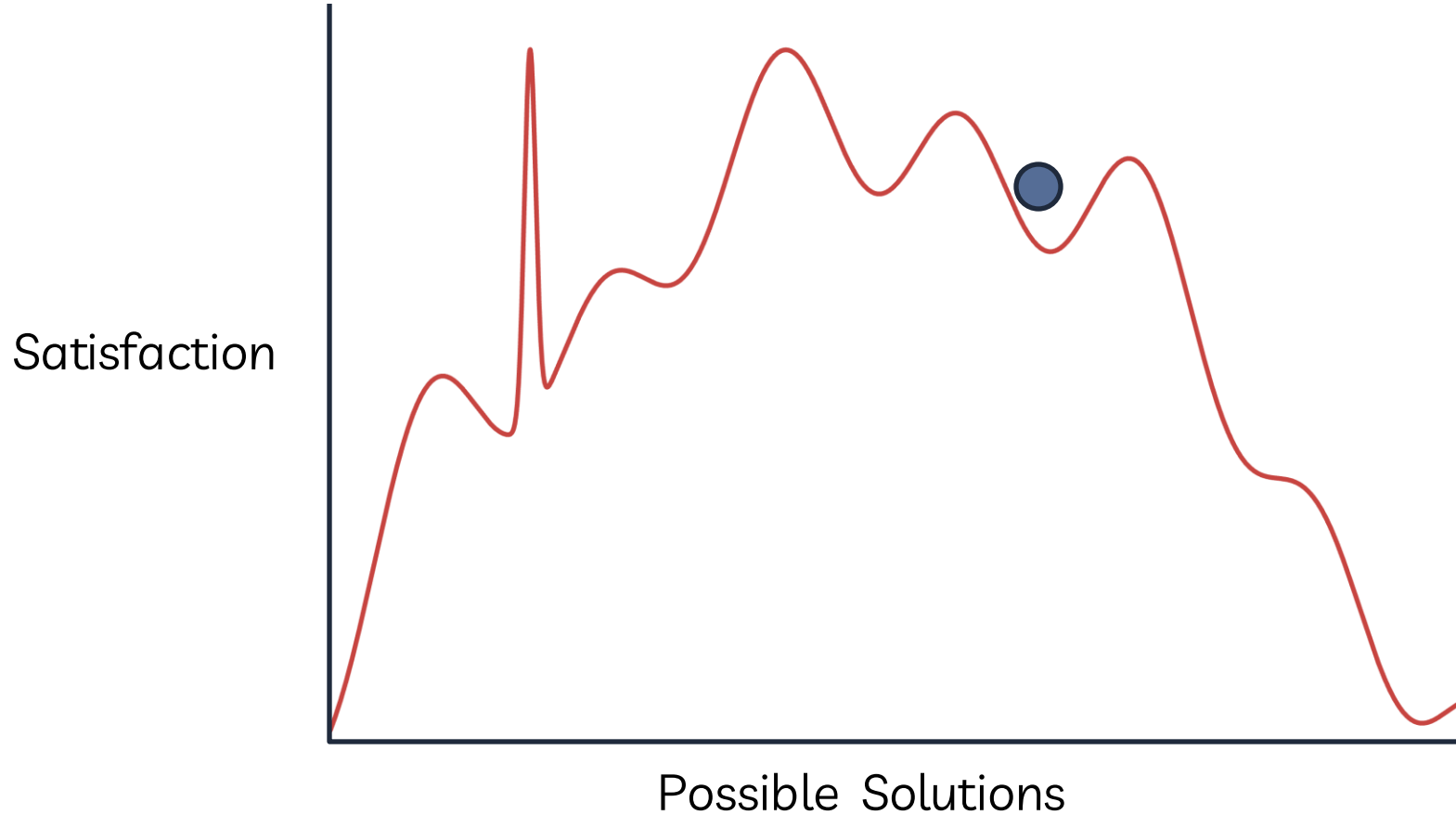
Non-Ideal: 7.8/10



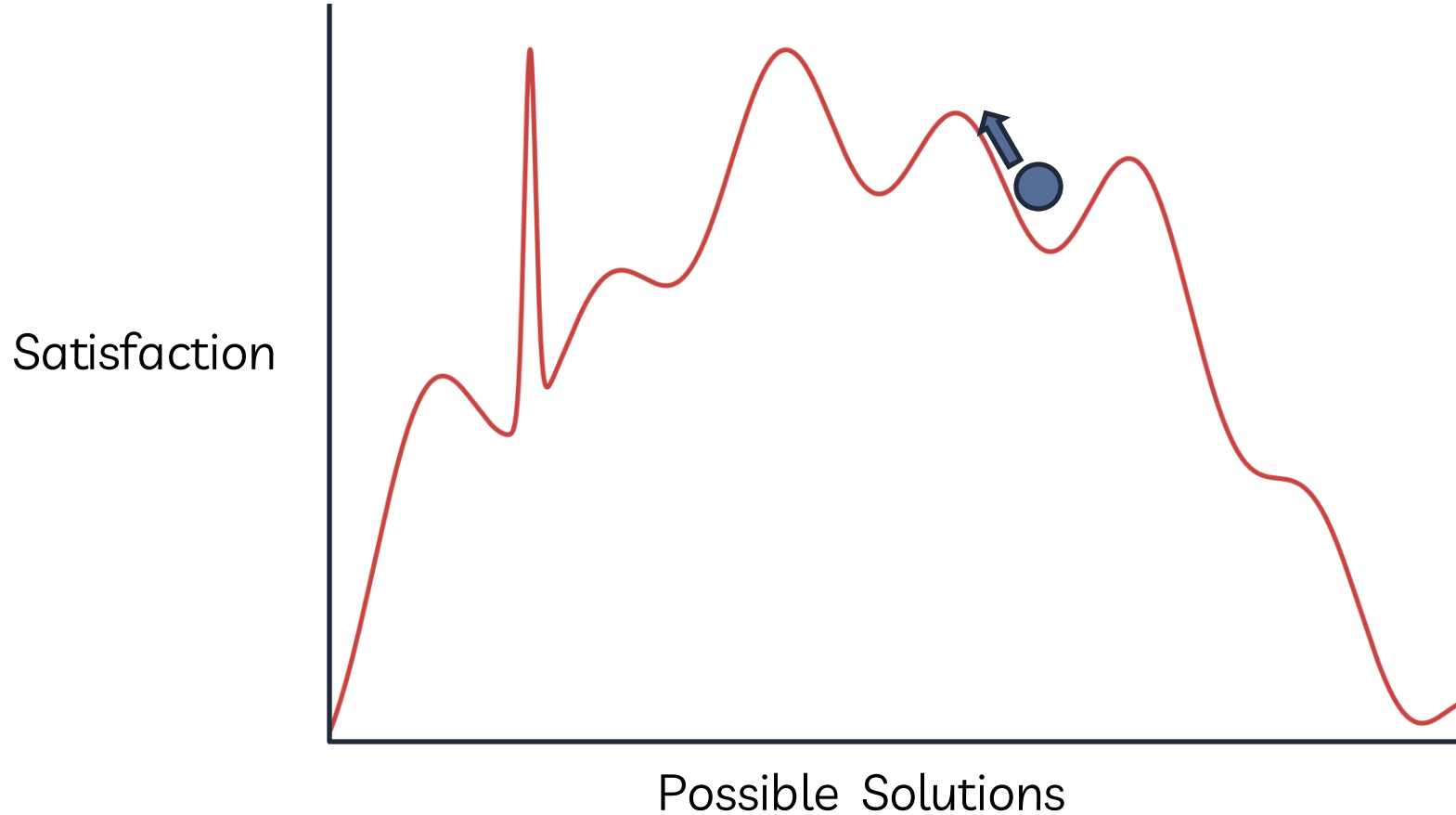
# Search Methods



# Search Methods: Depth Search



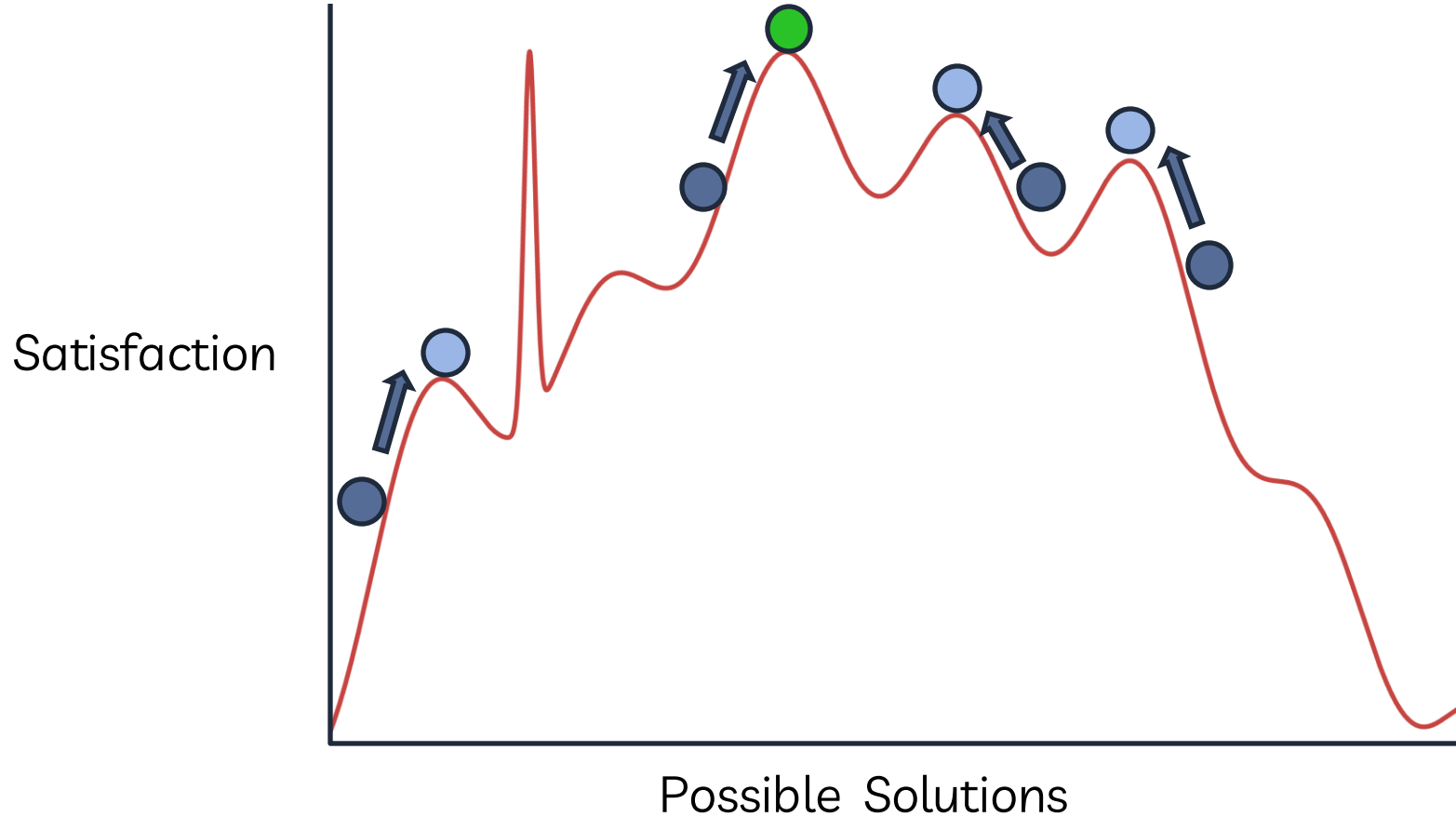
# Search Methods: Depth Search



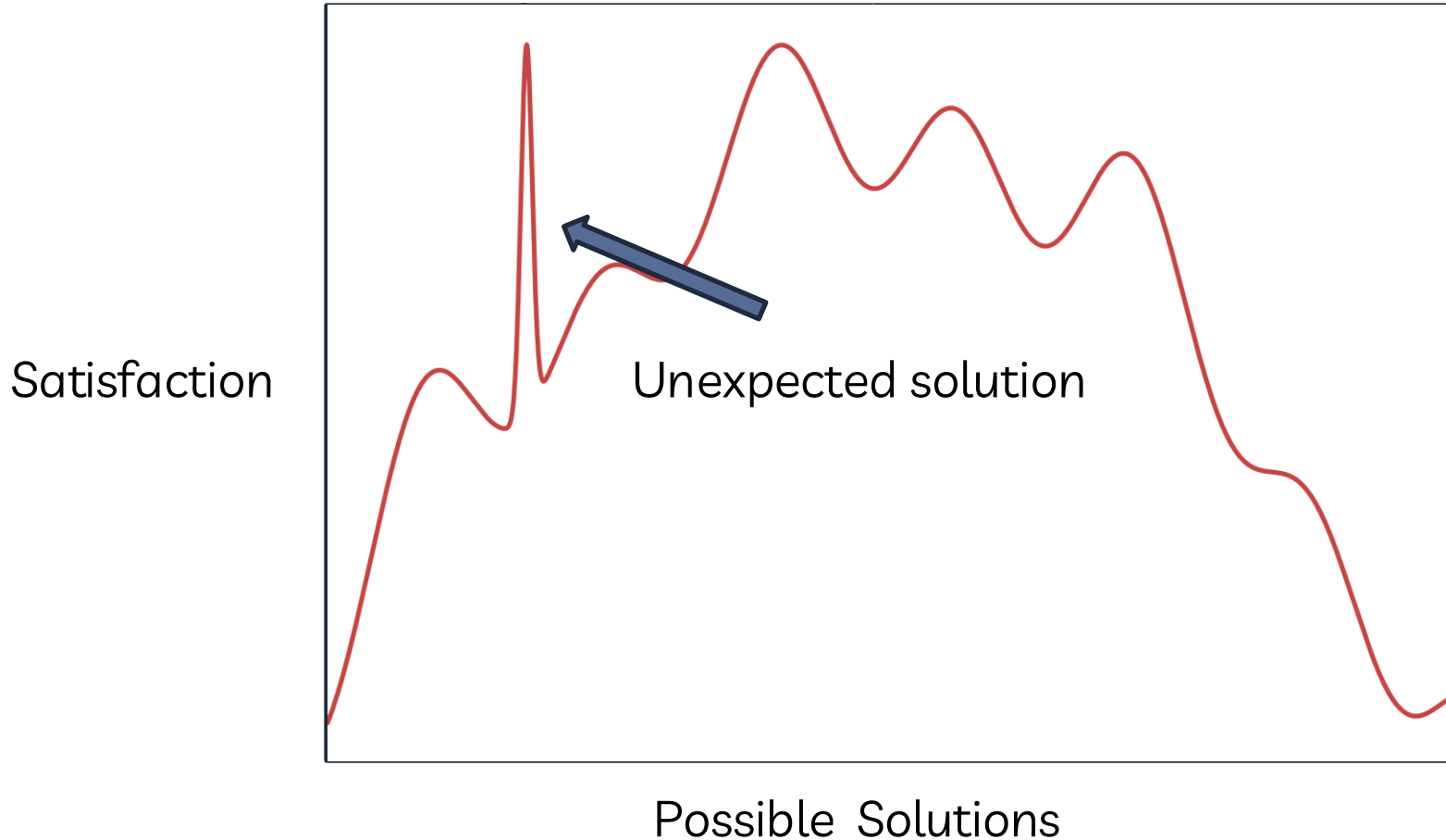
## Possible Solutions



# Search Methods: Depth Search



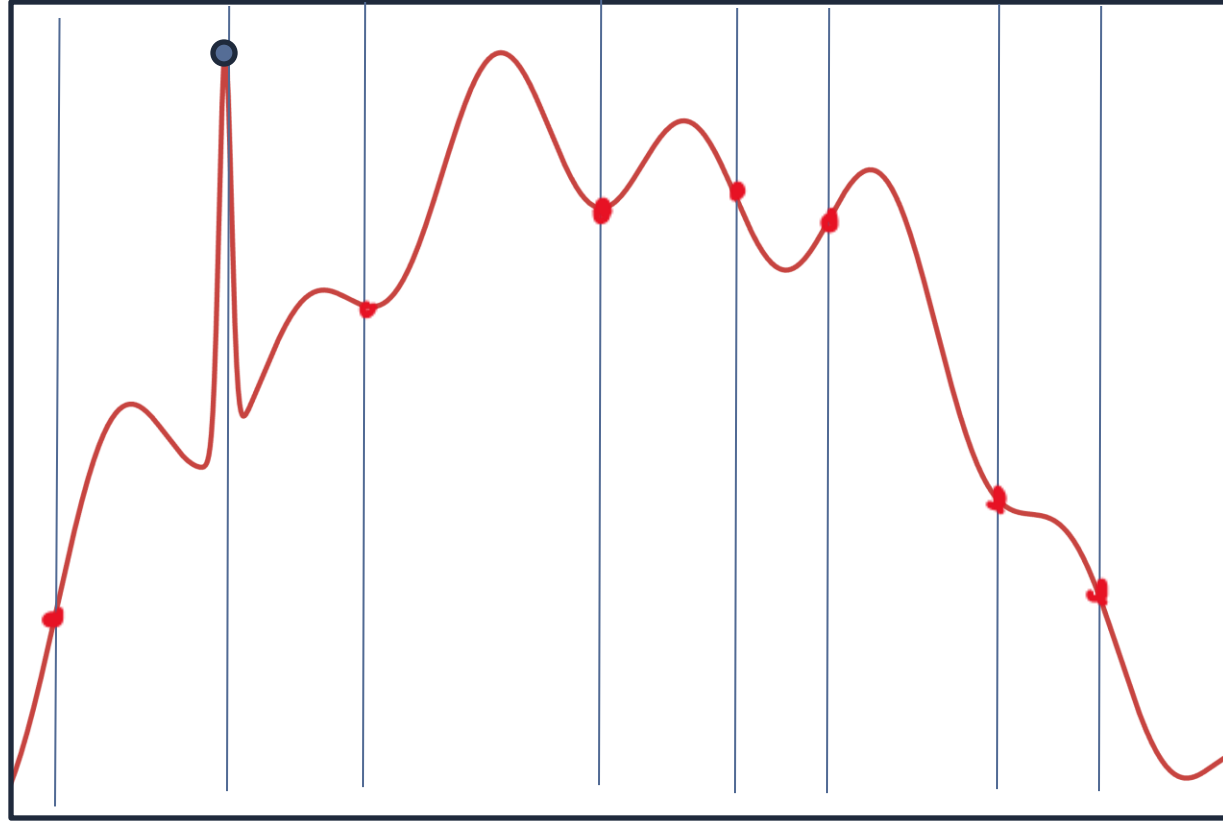
# Search Methods: Breadth Based



# Search Methods: Breadth Based

500 Random Samples

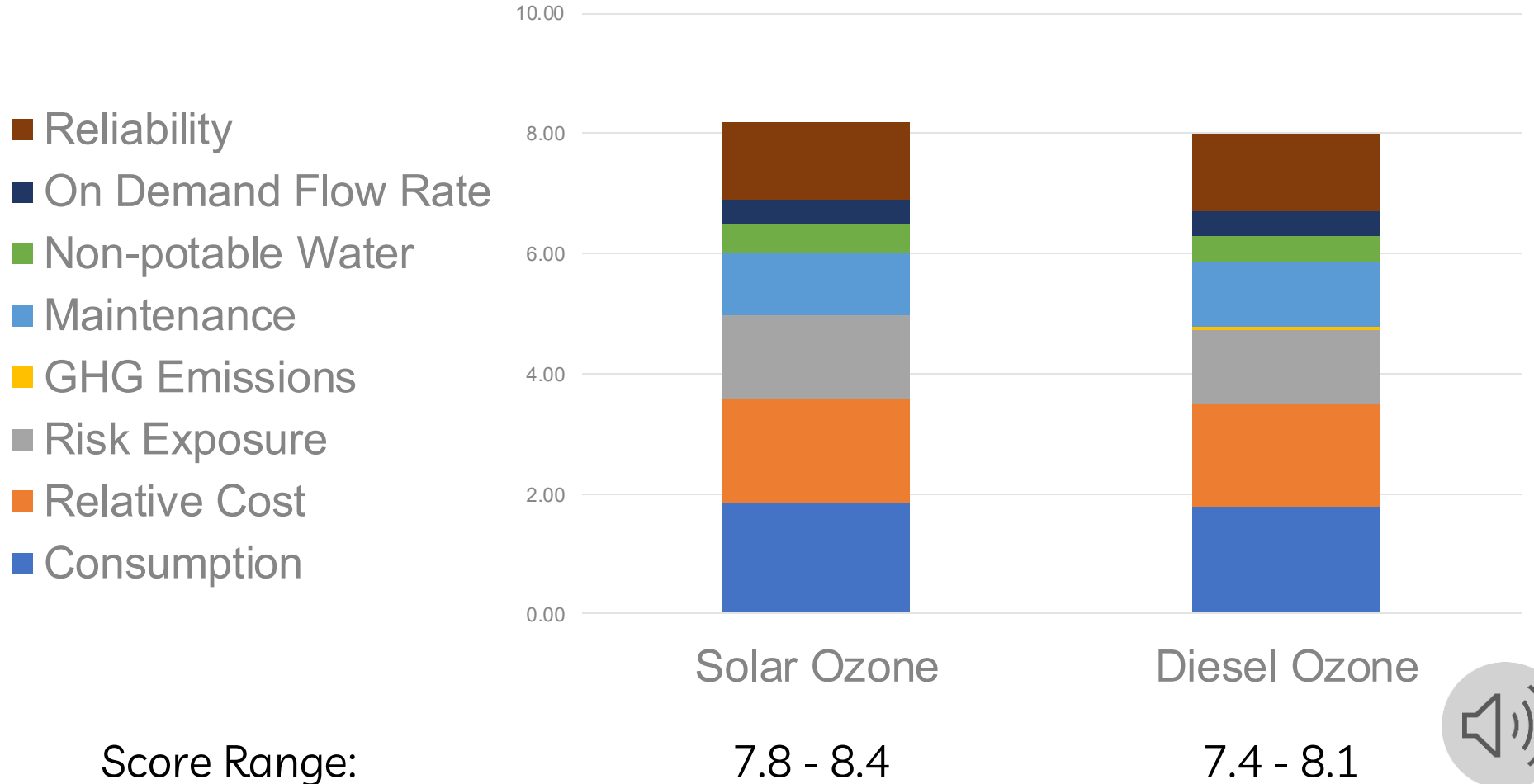
Satisfaction



Possible Solutions

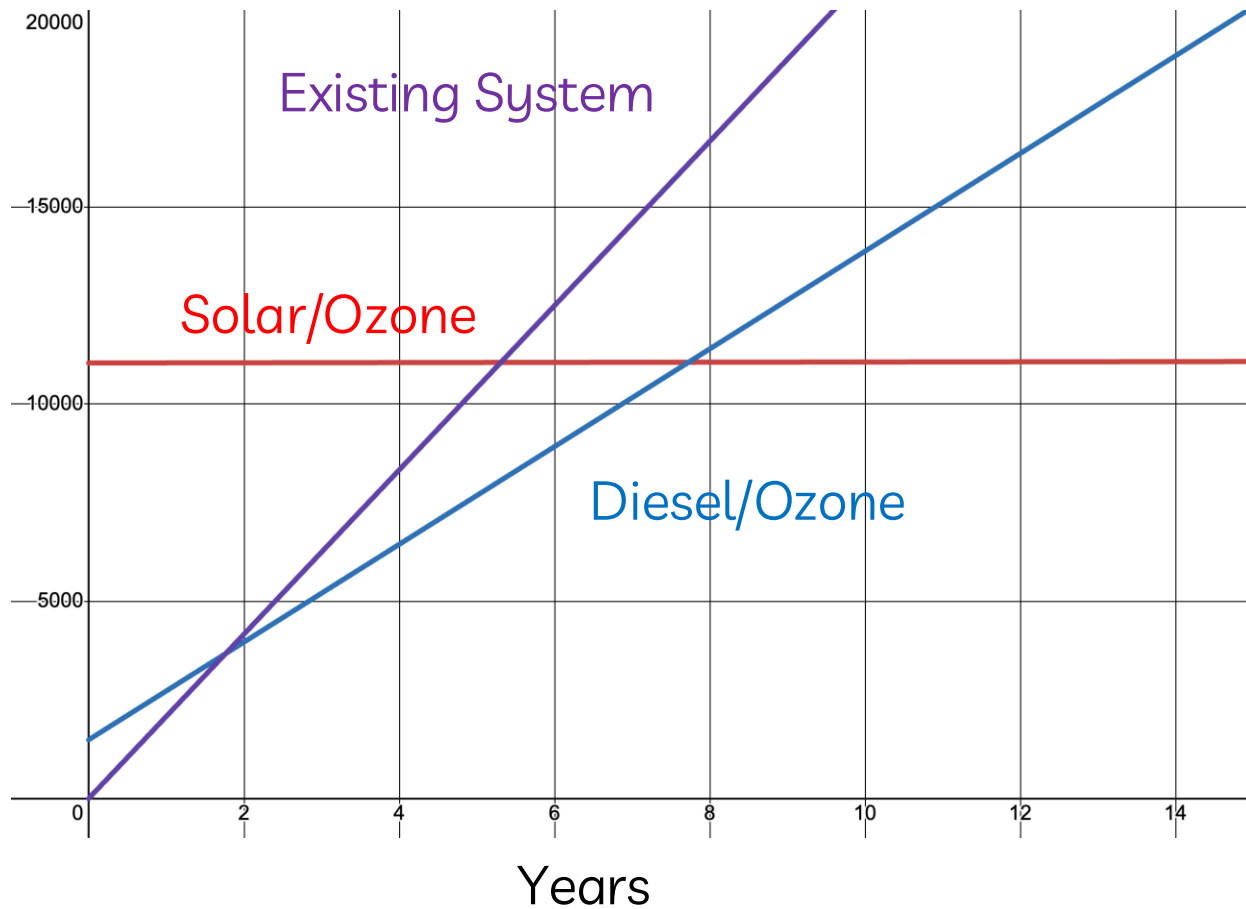


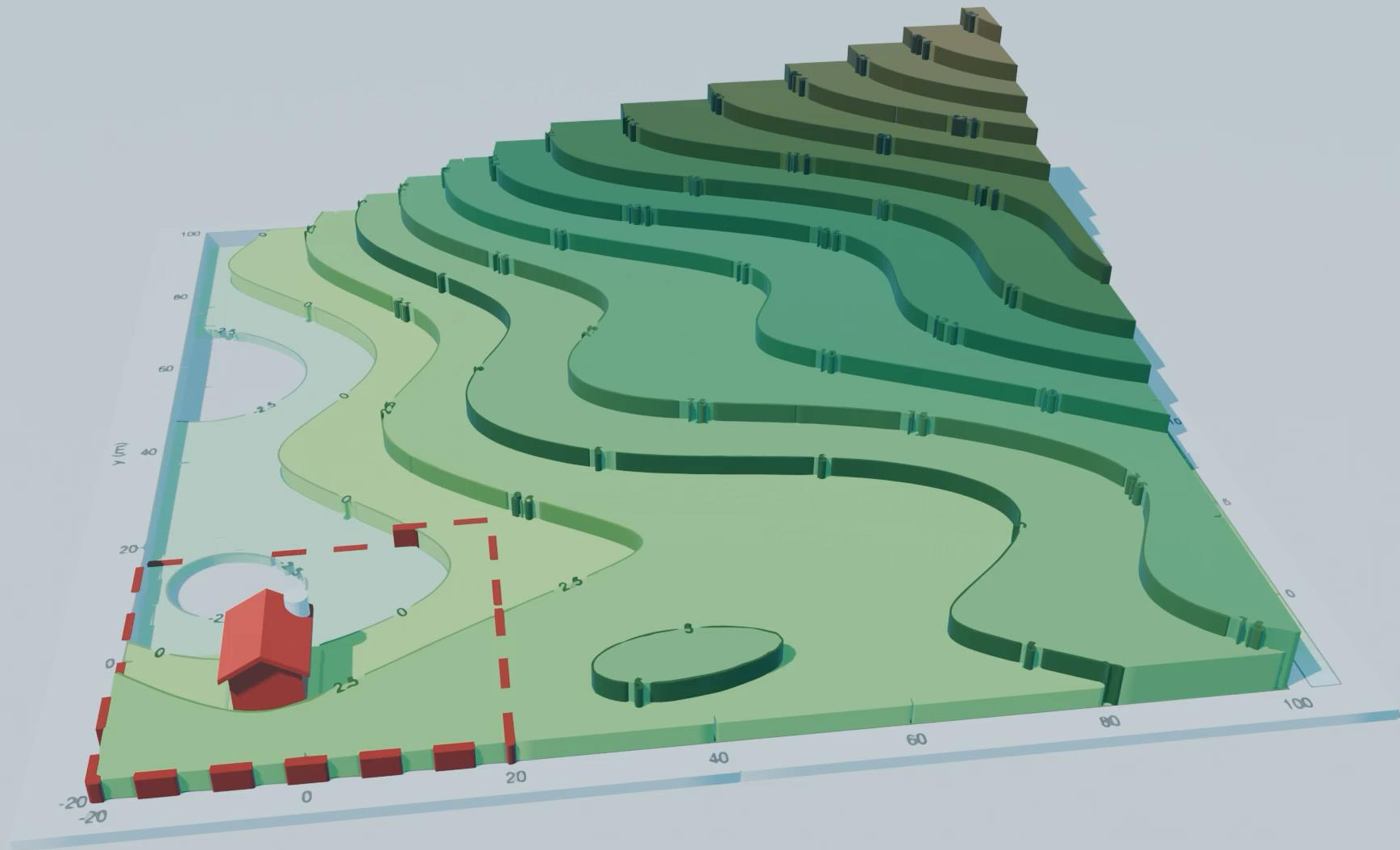
# WDM Score Comparison



# Long Term GHG Emissions

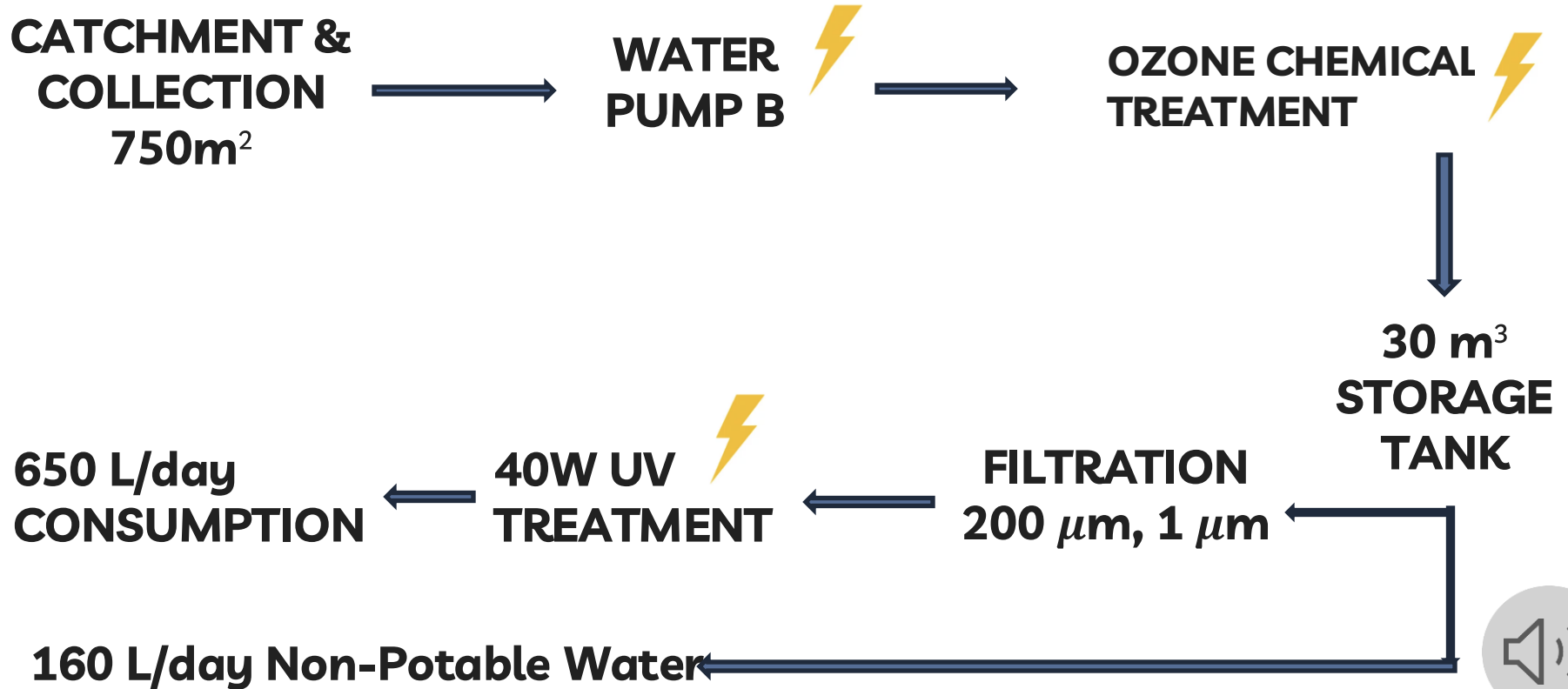
GHG Emissions  
(kgCO<sub>2</sub>e)





# FINAL DESIGN ELEMENTS

Power: 13 Solar Panels + 2 Batteries

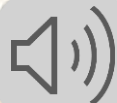




THANK YOU!

The background is a stylized map of the Strait of Georgia. A dark blue rectangular overlay covers the central part of the map. The text 'THANK YOU!' is written in white, bold, sans-serif capital letters in the center of this overlay. The map shows the coastline of the Strait of Georgia, with labels for VAN ANDA, TEXADA ISLAND, LASQUETI, SECHLT, Gibsons, Bowen, Gambier, and VANCOUVER. A small airplane icon is visible in the upper right. A scale bar at the bottom right indicates a distance of 100 km (62 miles).

\*Approximate distance  
100 km (62 miles)



# Generative AI / Credits

- Google Gemini was used to generate a map of Van Anda.
- Claude AI was used to aid in debugging a python script to random sample different configurations.
- ChatGPT was used to paraphrase bullet points for the relative cost slide.
- Images and spreadsheet data taken from APSC Component Selection Guide.
- Graphs and Spreadsheets from Microsoft Excel.
- 3D Topological Map prepared by Evan Fielding

